

Controlled Evaluation of Attention, Loss, and Upsampling Modifications in YOLOv10m for Packaging Defect Detection

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Abstract—Manual visual inspection of product packaging is time-consuming and may be affected by human error, especially when defect samples are limited and unevenly distributed. This study evaluates two controlled groups of YOLOv10m modifications for packaging-defect detection. Study I examines online Mosaic and MixUp augmentation, a sequential squeeze-and-excitation-coordinate-attention network, Balance IoU, and an adapted Cascade Confidence Loss. Study II evaluates RFACnv, Coordinate Attention, and CARAFE as candidate architectural modifications transferred from an improved YOLOv10 method. All configurations were initialized from the same pretrained checkpoint, trained at 640×640 pixels using three random seeds, and evaluated on the same 183-image held-out test set. The augmentation-only A1 configuration achieved the highest mAP_{50:95} of $57.89 \pm 1.59\%$. The original A0 architecture obtained $56.87 \pm 0.45\%$ and was the fastest configuration within Study II at 6.23 ± 0.02 ms/image. The complete E1 and E2 configurations reached $57.21 \pm 1.95\%$ and $54.93 \pm 0.46\%$, respectively. E2 also required 12.10 ± 0.03 ms/image. The results show that online augmentation provided the clearest improvement, whereas the transferred architectural modules did not improve held-out performance on this dataset.

Index Terms—packaging defect detection, YOLOv10m, data augmentation, attention mechanism, RFACnv, CARAFE

I. INTRODUCTION

Product packaging protects goods, communicates information, and influences consumer decisions. Ensuring packaging quality is therefore important for consumer safety and clear product communication, making quality control (QC) a fundamental manufacturing requirement. Traditionally, QC relies on manual visual inspection, which is slow, labor-intensive, and susceptible to human error.

With the advancement of deep learning (DL), Convolutional Neural Network (CNN)-based object detectors such as You Only Look Once (YOLO) have demonstrated strong potential for automating packaging-defect inspection. YOLO models perform object localization and classification within a single-stage detection framework. However, their performance depends strongly on the quality, quantity, and class distribution of the training data. Unequal class frequencies and limited defect samples may therefore reduce detection reliability.

Modern manufacturing lines are designed to minimize production errors, meaning that defective samples are naturally less common than normal products. This condition makes it difficult to collect sufficient defect images for model development [1]. A systematic review by de Giorgio et al. [2], covering 74 manufacturing studies, further identified class imbalance as a major limitation in industrial deep-learning applications.

Several strategies have been introduced to improve learning from limited industrial datasets. Data augmentation increases visual diversity through transformations or image mixing [3]. Attention mechanisms emphasize informative feature channels and spatial regions [4]. Modified loss functions adjust the optimization objective to improve localization or reduce the influence of easy samples [5]. Feature-extraction and upsampling modules have also been used to improve small-object representation. However, these modifications do not always transfer successfully between datasets and detector scales.

This study evaluates two controlled groups of YOLOv10m modifications on the Red Bull Can Appearance Defects Synthetic Dataset (RBCAD). Study I adapts the augmentation, multi-attention, and Balance Loss strategy introduced by Niu et al. [6]. Study II evaluates the cumulative RFACnv, Coordinate Attention, and CARAFE design reported by Wen et al. [11] as a candidate architecture transfer. The purpose is to identify the best configuration and determine which changes remain effective under the same split, pretrained checkpoint, training protocol, and held-out evaluation.

The key contributions of this study are as follows:

- MAnet, Balance IoU, and an adapted Cascade Confidence Loss are implemented for the one-to-many and one-to-one YOLOv10m training branches.
- RFACnv, Coordinate Attention, and CARAFE are transferred to YOLOv10m through a cumulative four-configuration architecture study.
- Detection accuracy, class behavior, model size, and inference latency are compared without assuming that an added module is an improvement.

II. RELATED WORK

Recent years have seen significant progress in applying deep-learning methods to product-packaging defect detection. Niu et al. [6] proposed an improved YOLOv4 model by optimizing data augmentation, attention, and loss backpropagation. Their method combined Mosaic and MixUp, introduced MANet by sequentially combining SENet channel attention with a modified coordinate-attention block, and replaced the original regression objective with Balance IoU and Cascade Confidence Loss. The complete model increased mAP by 1.2%. The study also showed that an attention block used alone did not always improve performance, which supports the need for controlled ablation.

Li et al. [7] introduced AS-YOLO based on YOLOv8 by combining synthetic-data generation, c2fCBAM modules, and Inner-IoU. Li et al. [8] proposed FSNB-YOLOv8 for surface-defect inspection using interactive augmentation, Normalized Wasserstein Distance loss, a modified backbone, and weighted BiFPN fusion. Cai et al. [9] developed PC-YOLO for ham-sausage inspection using image transformation, K-means clustering, Partial Convolution, and FPN-PAN fusion. Xiang et al. [10] proposed IM-YOLO for tiny paper-container defects using Multi-Scale Shuffle Attention and an Internal Feature Enhancement Network.

Wen et al. [11] improved YOLOv10n for dense remote-sensing detection by replacing selected backbone convolutions with RFACnv, replacing the post-SPPF attention block with Coordinate Attention, and replacing nearest-neighbor upsampling with CARAFE. Their complete model reported 97.2% mAP₅₀ and 63.2% mAP_{50:95} on UCAS-AOD. This architecture motivated Study II. Its transfer to YOLOv10m and packaging-defect data, however, requires direct evaluation because the object scale, image distribution, and detector capacity differ from the original task.

These studies show that augmentation, attention, loss modification, and feature reassembly can improve industrial or small-object detection. The remaining question is whether the same components provide stable gains when transferred to YOLOv10m under a fixed three-seed protocol. This study addresses that question using two controlled experiments on RBCAD.

III. DATA DESCRIPTION

The Red Bull Can Appearance Defects Synthetic Dataset (RBCAD), published by NeuroBot on Kaggle [15], is used in this study. The dataset contains 1,828 RGB images across 15 defect categories: black dot, black texture, bump, curved blade, dirty, green, lacerate, miss printing, missing bar code, molt, pinstripe, powder, red, transparent, and white. Each image has a resolution of 1080×1440 pixels and contains one annotated defect.

The raw images were stored by class and randomly split at image level while preserving the class ratio. The final split contains 1,279 training images, 366 validation images, and 183 test images, corresponding approximately to 70%, 20%, and 10%. The same split is used for all configurations. The

test set was not used for model selection, threshold selection, or training decisions; therefore, it represents an unseen-image test set from the same dataset distribution.

IV. METHODOLOGY

This section describes the YOLOv10m baseline and the components evaluated in the two controlled studies. Study I examines augmentation, attention, and loss changes. Study II examines a cumulative architecture transfer based on RFAConv, Coordinate Attention, and CARAFE.

A. Baseline YOLOv10m Architecture

YOLOv10m was selected as the baseline detector due to its balance between detection accuracy and computational efficiency [16]. The backbone uses Conv, C2f, SCDwn, C2fCIB, SPPF, and Partial Self-Attention (PSA) modules. The neck performs top-down and bottom-up feature fusion and produces P3, P4, and P5 feature maps. The YOLOv10m head uses one-to-many and one-to-one assignments during training, while the one-to-one branch supports end-to-end inference without non-maximum suppression [17].

Figs. 1–3 show the three architectures used in this work. Fig. 1 presents the original A0 architecture. Fig. 2 presents the complete E1 configuration, including the training-only augmentation and loss changes. Fig. 3 presents the complete E2 configuration, where RFACnv, Coordinate Attention, and CARAFE replace the corresponding original modules. The YOLOv10m head remains unchanged in all three architectures.

B. Study I: Augmentation, Attention, and Loss

1) *Mosaic and MixUp Augmentation*: The augmentation experiment compares a non-augmented baseline with a strong online augmentation recipe. For B0, HSV changes, geometric transformations, flipping, Mosaic, MixUp, CutMix, and copy-paste were disabled. For A1 and all later Study I configurations, the settings were $\text{hsv_h} = 0.015$, $\text{hsv_s} = 0.7$, $\text{hsv_v} = 0.4$, $\text{translate} = 0.1$, $\text{scale} = 0.5$, $\text{fliplr} = 0.5$, $\text{mosaic} = 1.0$, and $\text{mixup} = 1.0$. Rotation, shear, perspective transformation, vertical flipping, CutMix, and copy-paste remained disabled. Mosaic was closed during the final 10 epochs. A2–E1 inherit this same augmentation recipe so that the later comparisons isolate attention and loss changes.

2) *Multi-Attention Network*: MANet follows the sequential channel–spatial attention formulation of Niu et al. [6]. For a channel feature $u_{\text{channel}} \in \mathbb{R}^{H \times W}$, the squeeze operation is written in the reference as

$$\begin{aligned} z_{\text{channel}} &= F_{\text{squeeze}}(u_{\text{channel}}) \\ &= \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_{\text{channel}}(i, j), \quad z \in \mathbb{R}^C. \end{aligned} \quad (1)$$

The excitation operation calculates the channel weight and the channel-refined feature as

$$w = \text{Sigmoid}(W_2(\text{ReLU}(W_1 z_{\text{channel}}))), \quad s_{\text{channel}} = w \cdot u_c. \quad (2)$$

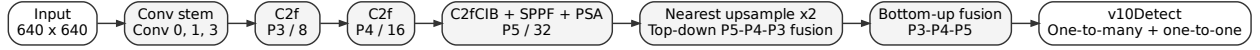


Fig. 1. Original YOLOv10m architecture used as the A0 baseline.

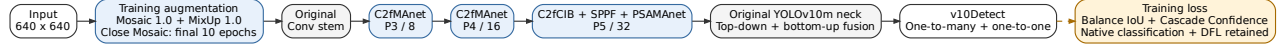


Fig. 2. Complete E1 architecture and training framework. MANet is inserted at P3/8 and P4/16, PSAMANet is used at P5/32, and the original YOLOv10m neck and detection head are retained.

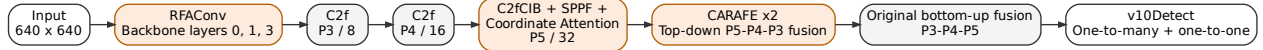


Fig. 3. Complete E2 architecture. RFACONV replaces backbone Conv layers 0, 1, and 3, Coordinate Attention replaces the post-SPPF PSA module, and CARAFE replaces both nearest-neighbor upsampling layers.

The modified coordinate-attention stage then pools s_{channel} along the two spatial directions:

$$\begin{aligned} z_{\text{channel}}^h(h) &= \frac{1}{W} \sum_{0 \leq i \leq W} s_{\text{channel}}(h, i), \\ z_{\text{channel}}^w(w) &= \frac{1}{H} \sum_{0 \leq j \leq H} s_{\text{channel}}(j, w). \end{aligned} \quad (3)$$

After concatenation, a shared 1×1 convolution, Layer Normalization, and separation into horizontal and vertical branches, the final MANet output is

$$y_c(i, j) = s_c(i, j) \times g_c^h(i) \times g_c^w(j). \quad (4)$$

Thus, channel attention is applied first and spatial attention is applied second. In this work, MANet is embedded through C2fMANet modules at P3/8 and P4/16 and a PSAMANet module at P5/32.

3) *Balance IoU and Cascade Confidence Loss*: The Balance IoU formulation in [6] combines EIou and CIou using an equilibrium coefficient:

$$\mathcal{L}_{\text{Balance IoU}} = \beta \mathcal{L}_{\text{EIou}} + (1 - \beta) \mathcal{L}_{\text{CIou}}, \quad (5)$$

$$\beta = \frac{e^{\mathcal{L}_{\text{EIou}}}}{e^{\mathcal{L}_{\text{EIou}}} + e^{\mathcal{L}_{\text{CIou}}}}. \quad (6)$$

This weighting uses more of the EIou term when its loss is larger and more of the CIou term when CIou is larger, providing a balance between early convergence and final localization accuracy. The important Cascade Confidence Loss expression from the reference is

$$\begin{aligned} \mathcal{L}_{\text{Cascade confidence}} &= \mathcal{L}_{\text{bce}} + \alpha_1 \lambda_{\text{ignore}}^1 \mathcal{L}_{\text{bce}}^{\text{noobj}} \\ &\quad + \alpha_2 \lambda_{\text{ignore}}^2 \mathcal{L}_{\text{bce}}^{\text{noobj}} + \alpha_3 \lambda_{\text{ignore}}^3 \mathcal{L}_{\text{bce}}^{\text{noobj}}. \end{aligned} \quad (7)$$

The three IoU thresholds are 0.5, 0.6, and 0.7, while $(\alpha_1, \alpha_2, \alpha_3) = (1, 0.5, 0.25)$. YOLOv10m has no independent objectness output; therefore, the same cascade thresholds and weights were adapted to the assigned-class confidence logits of foreground predictions in both training branches. The native classification and Distribution Focal Loss components remain active.

C. Study II: Transferred Architecture Modules

Study II follows the cumulative design of Wen et al. [11], but applies it to YOLOv10m and packaging-defect images. The equations below retain the main formulations used in the reference paper, while only the modules specified by C1, C2, and E2 are changed.

1) *Receptive Field Attention Convolution*: C1 replaces backbone Conv layers 0, 1, and 3 with RFACONV. Following the notation in [11], the receptive-field attention weight is

$$w = \text{Softmax}(g^{1 \times 1}(\text{AvgPool}(x))), \quad (8)$$

and the receptive-field feature is

$$x' = \text{ReLU}(\text{Norm}(g^{k \times k}(x))). \quad (9)$$

The output is then obtained by weighting the receptive-field feature:

$$y = w \times x'. \quad (10)$$

Here, $g^{1 \times 1}$ and $g^{k \times k}$ denote grouped convolutions, and $k = 3$ is used in this experiment. The operation allows the response within each local receptive field to depend on the feature importance at that position.

2) *Coordinate Attention*: C2 inherits C1 and replaces the PSA module after SPPF with Coordinate Attention. The reference first forms horizontal and vertical direction-aware features:

$$\begin{aligned} x^h &= \frac{1}{W} \sum_{0 \leq i \leq W} x(h, i), \\ x^w &= \frac{1}{H} \sum_{0 \leq j \leq H} x(j, w). \end{aligned} \quad (11)$$

They are concatenated and transformed as

$$z = \delta(F_1([x^h, x^w])), \quad (12)$$

then separated into z^h and z^w to obtain

$$\begin{aligned} a^h &= \sigma(F_h(z^h)), & a^w &= \sigma(F_w(z^w)), \\ y &= x \times a^h \times a^w. \end{aligned} \quad (13)$$

Here, F_1 , F_h , and F_w are 1×1 convolution operations, δ is a nonlinear activation, and σ is the sigmoid function. The module keeps horizontal and vertical position information while reweighting the input feature.

3) *Content-Aware Reassembly of Features*: E2 inherits C2 and replaces both factor-two nearest-neighbor upsampling layers in the neck with CARAFE. The module first predicts and normalizes a content-aware reassembly kernel. For a local window of size $k_{up} \times k_{up}$, the reference expresses the reassembled output as

$$y(i, j) = \sum_{m=0}^{k_{up}-1} \sum_{n=0}^{k_{up}-1} w(m, n) \cdot x(i+m, j+n). \quad (14)$$

The weights $w(m, n)$ are generated from the local feature content and normalized so that the kernel weights sum to one. In E2, the scale factor is 2, the compressed channel width is 64, the content encoder is 3×3 , and $k_{up} = 5$. Group-1 reassembly is processed in channel chunks of 32 to reduce peak memory without changing the operation.

V. EXPERIMENTAL SETUP

Both studies were implemented using a locally frozen Ultralytics 8.3.247 codebase with Python 3.9.21 and PyTorch 2.5.1+cu121. Training and testing were executed on an NVIDIA GeForce RTX 2080 Ti using CUDA 12.1. Every configuration was initialized from the same official YOLOv10m checkpoint.

All models used 640×640 input images, 100 epochs, a physical batch size of 16, gradient accumulation of four, eight workers, automatic mixed precision, and deterministic execution. Ultralytics optimizer=auto selected AdamW with an effective initial learning rate of approximately 5.26×10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay of 0.0005. Cosine scheduling was disabled. Seeds 42, 123, and 2026 were used for every configuration.

Study I compares an augmentation-disabled baseline with the configurations in Table I. B0 disables train-time augmentation. A1 activates Mosaic and MixUp. A2–A4 add SE,

TABLE I
STUDY I CONFIGURATIONS

ID	Augmentation	Attention	Regression	Confidence
B0	Disabled	None	Native	Native
A1	Mosaic + MixUp	None	Native	Native
A2	Mosaic + MixUp	SE	Native	Native
A3	Mosaic + MixUp	modified CA	Native	Native
A4	Mosaic + MixUp	MANet	Native	Native
L1	Mosaic + MixUp	MANet	EIoU	Native
L2	Mosaic + MixUp	MANet	Balance IoU	Native
E1	Mosaic + MixUp	MANet	Balance IoU	Cascade

TABLE II
STUDY II CONFIGURATIONS

ID	Backbone	Post-SPPF	Upsampling	Head
A0	Original Conv	PSA	Nearest neighbor	Native
C1	RFACnv at 0, 1, 3	PSA	Nearest neighbor	Native
C2	RFACnv at 0, 1, 3	Coord. Attention	Nearest neighbor	Native
E2	RFACnv at 0, 1, 3	Coord. Attention	CARAFE	Native

modified CA, or MANet. L1 and L2 use EIoU or Balance IoU, while E1 adds Cascade Confidence Loss to L2.

Study II uses Mosaic 1.0, MixUp 0.0, horizontal flipping 0.5, translation 0.1, and scale 0.5. Mosaic is disabled during the final 10 epochs. Its cumulative configurations are listed in Table II.

For each run, `best.pt` was selected using validation $mAP_{50:95}$ only. The fixed 183-image test split was evaluated after all checkpoints were complete. No model was retrained, reselected, or tuned after the test results were obtained.

Precision, recall, and F1-score are calculated as

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R}. \quad (15)$$

The mAP_{50} metric uses an IoU threshold of 0.50, while $mAP_{50:95}$ averages AP from 0.50 to 0.95 in steps of 0.05. Parameter count, GFLOPs, and inference latency are also reported. Metrics are shown as mean $\pm$ sample standard deviation across three seeds. The two studies were produced by separate locked experiment packages; therefore, cross-study latency and GFLOP values are descriptive, while controlled efficiency comparisons are made mainly within each study.

VI. RESULTS AND DISCUSSION

A. Combined Held-Out Comparison

Table III combines the held-out results of both studies. Across both studies, A1 achieved the highest $mAP_{50:95}$ at $57.89 \pm 1.59\%$. A0 achieved the highest precision, while L1 achieved the highest recall and mAP_{50} . The complete E1 configuration reached $57.21 \pm 1.95\%$, which was 2.28 percentage points above E2. This difference is descriptive because E1 and E2 belong to different controlled paths.

B. Study I: Augmentation, Attention, and Loss

B0 achieved $53.11 \pm 1.53\%$ $mAP_{50:95}$. Activating the online augmentation recipe in A1 increased the result by

TABLE III

HELD-OUT PERFORMANCE ACROSS THREE SEEDS (MEAN $\pm$ SAMPLE STANDARD DEVIATION). Δ IS MEASURED FROM THE BASELINE OF EACH STUDY.

Study	ID	P (%)	R (%)	F1 (%)	mAP ₅₀ (%)	mAP _{50:95} (%)	Δ (pp)
I	B0	89.23 $\pm$ 2.96	82.02 $\pm$ 3.56	85.40 $\pm$ 1.50	89.69 $\pm$ 1.90	53.11 $\pm$ 1.53	–
I	A1	91.94 $\pm$ 1.49	92.13 $\pm$ 1.41	92.03 $\pm$ 1.30	95.13 $\pm$ 1.07	57.89 $\pm$ 1.59	+4.78
I	A2	90.26 $\pm$ 1.55	91.36 $\pm$ 3.03	90.80 $\pm$ 2.18	93.72 $\pm$ 1.78	57.66 $\pm$ 0.66	+4.55
I	A3	90.72 $\pm$ 0.37	91.80 $\pm$ 1.82	91.25 $\pm$ 0.77	94.53 $\pm$ 0.98	57.40 $\pm$ 0.13	+4.29
I	A4	91.54 $\pm$ 1.95	90.72 $\pm$ 1.34	91.12 $\pm$ 1.06	94.21 $\pm$ 1.45	57.10 $\pm$ 1.13	+3.99
I	L1	91.16 $\pm$ 4.38	92.81 $\pm$ 0.70	91.94 $\pm$ 2.11	95.34 $\pm$ 0.79	57.24 $\pm$ 0.79	+4.13
I	L2	90.44 $\pm$ 3.19	91.64 $\pm$ 1.07	91.02 $\pm$ 1.88	93.83 $\pm$ 1.28	56.19 $\pm$ 1.02	+3.08
I	E1	90.81 $\pm$ 3.68	90.84 $\pm$ 0.17	90.80 $\pm$ 1.84	93.99 $\pm$ 0.70	57.21 $\pm$ 1.95	+4.10
II	A0	92.24 $\pm$ 1.79	90.47 $\pm$ 2.70	91.32 $\pm$ 0.80	93.50 $\pm$ 0.89	56.87 $\pm$ 0.45	–
II	C1	90.19 $\pm$ 3.97	90.89 $\pm$ 1.13	90.49 $\pm$ 1.43	93.37 $\pm$ 0.32	55.16 $\pm$ 0.89	-1.71
II	C2	89.98 $\pm$ 3.27	87.88 $\pm$ 0.80	88.89 $\pm$ 1.17	92.14 $\pm$ 0.46	54.86 $\pm$ 1.04	-2.00
II	E2	90.49 $\pm$ 1.52	88.46 $\pm$ 2.11	89.45 $\pm$ 1.43	92.72 $\pm$ 0.55	54.93 $\pm$ 0.46	-1.94

4.78 percentage points and produced the best overall primary metric. The attention variants A2, A3, and A4 remained close to A1 but did not exceed it. A3 had the lowest variation for mAP_{50:95}, with a sample standard deviation of 0.13 percentage points.

L1 produced the highest recall of $92.81 \pm 0.70\%$ and mAP₅₀ of $95.34 \pm 0.79\%$, but its stricter mAP_{50:95} was below A1. Balance IoU reduced the mean result in L2, while Cascade Confidence Loss recovered part of this reduction in E1. Therefore, the main improvement in Study I came from augmentation, not from the complete E1 combination.

C. Study II: RFACnv, Coordinate Attention, and CARAFE

A0 obtained the best Study II result at $56.87 \pm 0.45\%$ mAP_{50:95}. C1 increased mean recall by 0.42 percentage points, but reduced the primary metric by 1.71 points. C2 reduced the parameter count by 7.30% relative to A0, but its primary metric was 2.00 points lower. E2 improved C2 by only 0.06 points on average, and the sign of the change differed across seeds. E2 remained 1.94 points below A0 and increased mean inference latency from 6.23 to 12.10 ms/image.

These results do not support describing E2 as an enhanced model for RBCAD. RFACnv, Coordinate Attention, and CARAFE were evaluated as enhancement candidates because they improved YOLOv10n on UCAS-AOD in [11]. On the present dataset, E2 is more accurately described as the complete transferred configuration. The result shows that an architecture effective for dense remote-sensing objects may not transfer directly to packaging defects or to the medium YOLOv10 scale.

D. Model Size and Latency

Table IV combines model size and latency for both studies. Study I changes introduced small parameter and GFLOP increases. A1 retained the baseline architecture and achieved the best primary metric in that study. In Study II, C2 and E2 were smaller than A0, but the custom operations did not improve runtime on the RTX 2080 Ti. The small GFLOP difference between the Study I and Study II baseline profiles is caused by the separate locked profiling pipelines and is not an architecture difference.

TABLE IV
MODEL SIZE AND HELD-OUT INFERENCE LATENCY.

Study	ID	Param. (M)	GFLOPs	Latency (ms)
I	B0	16.501	64.058	8.445 $\pm$ 0.054
I	A1	16.501	64.058	8.423 $\pm$ 0.076
I	A2	16.567	64.115	8.459 $\pm$ 0.050
I	A3	16.553	64.140	8.536 $\pm$ 0.085
I	A4	16.619	64.197	8.609 $\pm$ 0.048
I	L1	16.619	64.197	8.644 $\pm$ 0.005
I	L2	16.619	64.197	8.698 $\pm$ 0.060
I	E1	16.619	64.197	8.660 $\pm$ 0.077
II	A0	16.501	63.505	6.227 $\pm$ 0.023
II	C1	16.517	63.911	8.860 $\pm$ 0.050
II	C2	15.297	62.929	8.731 $\pm$ 0.023
II	E2	15.474	63.352	12.102 $\pm$ 0.029

VII. CONCLUSION

This study evaluated augmentation, attention, loss, and architecture modifications for YOLOv10m packaging-defect detection using two controlled experiments and three random seeds. In Study I, online Mosaic and MixUp provided the clearest improvement. A1 achieved the highest overall mAP_{50:95} of $57.89 \pm 1.59\%$, while the complete E1 configuration reached $57.21 \pm 1.95\%$. In Study II, the original A0 architecture achieved $56.87 \pm 0.45\%$ and the best accuracy–latency balance. The complete E2 configuration reached $54.93 \pm 0.46\%$ and almost doubled the inference latency relative to A0. Therefore, E2 should be treated as an evaluated architecture transfer rather than an achieved enhancement on RBCAD. The results show that added modules do not automatically improve generalization, and that controlled held-out evaluation is necessary before an architectural change is described as beneficial.

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